

Exercise 3: Stored Procedures

Scenario 1: The bank needs to process monthly interest for all savings accounts.

Question: Write a stored procedure **ProcessMonthlyInterest** that calculates and updates the balance of all savings accounts by applying an interest rate of 1% to the current balance.

Program

```
CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest
IS
BEGIN

    UPDATE Accounts

    SET Balance = Balance + (Balance * 0.01)

    WHERE AccountType = 'Savings';

    COMMIT;

    DBMS_OUTPUT.PUT_LINE('Monthly interest applied successfully.');
```

EXCEPTION

```
    WHEN OTHERS THEN

        ROLLBACK;

        DBMS_OUTPUT.PUT_LINE('Error: ' || SQLERRM);
```

END;

Scenario 2: The bank wants to implement a bonus scheme for employees based on their performance.

Question: Write a stored procedure **UpdateEmployeeBonus** that updates the salary of employees in a given department by adding a bonus percentage passed as a parameter.

Program

```
CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus(  
    p_department VARCHAR2,  
    p_bonus NUMBER  
)  
IS  
  
BEGIN  
  
    UPDATE Employees  
    SET Salary = Salary + (Salary * p_bonus / 100)  
    WHERE Department = p_department;  
  
    COMMIT;  
  
    DBMS_OUTPUT.PUT_LINE('Employee bonus updated successfully.');
```

EXCEPTION

WHEN OTHERS THEN

ROLLBACK;

DBMS_OUTPUT.PUT_LINE('Error: ' || SQLERRM);

END;

Scenario 3: Customers should be able to transfer funds between their accounts.

Question: Write a stored procedure **TransferFunds** that transfers a specified amount from one account to another, checking that the source account has sufficient balance before making the transfer.

Program

CREATE OR REPLACE PROCEDURE TransferFunds(

p_fromAccount NUMBER,

p_toAccount NUMBER,

p_amount NUMBER

)

IS

v_balance NUMBER;

BEGIN

SELECT Balance

INTO v_balance

FROM Accounts

WHERE AccountID = p_fromAccount;

IF v_balance >= p_amount THEN

UPDATE Accounts

SET Balance = Balance - p_amount

WHERE AccountID = p_fromAccount;

UPDATE Accounts

SET Balance = Balance + p_amount

WHERE AccountID = p_toAccount;

COMMIT;

DBMS_OUTPUT.PUT_LINE('Fund transfer completed successfully.');

ELSE

DBMS_OUTPUT.PUT_LINE('Insufficient balance. Transfer cancelled.');

END IF;

EXCEPTION

WHEN OTHERS THEN

ROLLBACK;

DBMS_OUTPUT.PUT_LINE('Error: ' || SQLERRM);

END;